2023 IEEE 3rd International Conference on Sustainable Energy and Future Electric Transportation (SEFET) | 979-8-3503-1997-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/SEFET57834.2023.10245825

# Real-Time Digital Simulation for Investigating Transient Stability of Power System Network

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***Abstract*—This paper examines the power swing explicitly before and after a fault, which plays a crucial role in stabilizing the power system network during transient conditions. It analyzes the transient stability of the IEEE 5 bus network during a three-phase symmetrical fault occurrence and load variation. The purpose is to limit voltage transients during the fault by selecting appropriate Critical Clearing Time (CCT) and breaker resistance. Additionally, the article investigates the impact of wind generator integration on the voltage and frequency improvement of the power system network during load variations. The study employs a Real-Time Digital Simulator (RTDS) and an open-source software Power World Simulator (PWS). Phasor Measurement Units (PMUs) are utilized to monitor the voltage and frequency of the network. The accuracy and effectiveness of the results obtained from the RTDS and PWS are compared to validate the power system transient stability analysis.**

***Index Terms*—Transient Analysis, Three Phase Fault, Real Time Digital Simulator (RTDS), Electromagnetic Transient (EMT), Critical Clearing Time (CCT), Wind Generation (WG), Transient Recovery Voltage (TRV)**

## I. Introduction

Transient stability analysis is an essential tool for power system engineers to ensure the reliable and secure operation of power systems. Transient stability is concerned with the ability of the power system to maintain synchronism when subjected to severe disturbances such as faults and sudden changes in load demand. Disturbances cause a mismatch in power generation and consumption and cause the system to lose synchronism along with the deviation in rotor angle of the generator, bus voltages, frequency, and power flow [1]. To ensure the reliable and secure operation of any power system, it is highly desirable that all the parameter's stability must be well preserved in the network. A slight departure of the operating point beyond some critical level might initiate voltage and frequency instability in the system. Instability due to faults can be stabilised using protective devices such as circuit breakers, relays etc. For the operation of circuit breaker it is important to find CCT. In the IEEE report, the CCT is defined as "the maximum time between the fault initiation and its clearing such that the power system is transiently stable". The CCT is an efficient factor for the estimation of transient stability it determines whether the system will regains stability after fault clearance or not.

In [2], author proposed various methods to calculate the correct time for functioning of circuit breaker after the occurrence of disturbance. Another contingency such as the load variation it cause voltage instability in the system and is treated as a local load bus problem.

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This voltage instability occurs due to a deficiency of reactive power during load variation. In [3], reason for the voltage instability and how to improve voltage stability are studied. As proposed in [4] voltage stability can be improved by adding reactive sources such as wind turbine. These transmission system disturbances also leads to frequency deviation within acceptable limits and other expected stability challenges as proposed in [5]. However, the real time operation of the protective device with calculated time is required to be simulated for analysing the stability of the system.

In this context, the present topic focuses on the transient stability analysis of a 5-bus network during a 3-phase fault. The objective is to investigate the behavior of the network under these conditions, including the operation of circuit breakers to clear the fault. Furthermore, the impact of load variation on network parameters is examined, and the integration of wind generation [6] is proposed to improve the system stability during load increments. The integration of wind generation is particularly relevant, given the increasing use of renewable energy sources in power systems. This topic aims to provide insights into the behavior of power systems during disturbances and the effectiveness of wind generation integration in improving system stability.

Most of the software used for power systems study is non-real-time, allowing users to study a specific change on a particular power system model over the desired period [7]. Some real time platforms such as OPAL-RT, Mi- power were used for the transient stability analysis in real time [8] [9]. In this paper, two simulators, Real-Time Digital Simulator (RTDS) and Power World Simulator (PWS) are used for analysis purposes so that results can be cross-verified [10]. Real-world operating scenarios can be generated using an RTDS simulator with Hardware-In-the-Loop (HIL) [11] [12].

The main aim of the investigation is as follows:

1) To perform a real-time analysis of the transient stability of the IEEE 5-bus network in response to a three-phase fault at bus 4.
2) To determine the CCT and resistance required for the breaker operation to limit voltage transients after the fault is recovered.
3) To investigate the frequency and voltage regulation of the system while integrating a wind generator under load variation.

The paper is organised as follows: Power flow analysis pre-fault and post fault described in section II . The brief description of the simulation tools are illustrated in section III. Section IV discusses the simulation results using for two



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cases one for Fault near bus 4 and second for WG integration to improve stability during load variation. The description is subsequently concluded in section VI.

## II. Power Flow Analysis

As shown in Fig. 1, the considered power system network comprises of two generating units of constant inertia with the step up transformers and an infinite generating source. Loads are connected at bus4 and bus5. For a well-functioning system, load flow analysis is necessary to detect possible errors. Various methods for load flow problems were introduced in [13]. There are different mathematical solutions for the load flow analysis problem and the most widely used are Newton-Raphson, Fast-Decoupled and Gauss-Seidel methods [14]. In this paper Newton Raphson method is used for the load flow analysis of the 5 bus power network before and after the fault along with the breaker operation. Transient stability analysis necessitates pre-fault power flow and provides the transmission line flow, bus voltages, and angles. Newton Raphson method has been performed for the load flow analysis method with hundreds of iterations to achieve the best results. This method employs preliminary guessing of unknown parameters. Bus admittance matrix ($Y_{bus}$) has been generated using generator and load reactance parameters from given tabular data in [15]

### A. Desired swing equation during fault

For the determination of the swing equation of the power system, transient internal voltage ($E_i{}'$) following the current ($I_i$) flowing from the generating sources into the network are measured as follows

$$I_i = \frac{(P_i + jQ_i)^*}{V_i^*} \qquad E_i^{'} = V_i + jX_d^{'}I_i \tag{1}$$

In the considered 5 bus network, two generators are connected at the Bus 1 and Bus 2 along with the infinite Bus 3. Therefore, $E_3 = 1\angle 0°$. The loads are connected at Bus 4 and Bus 5 are converted into the shunt admittance using $Y_l = \frac{(P_l - jQ_l)^*}{V_l^2}$

Bus admittance matrix before fault is modified to $5\times5$ $Y_{bus}$ matrices including load admittances and transient reactances of the machine. When the fault is introduced at transmission lines $4-5$ near Bus 4. The $5\times5$ $Y_{bus}$ matrices is reduced to $3\times3$ $Y_{bus}$ matrice by Kron reduction method [15].

Under fault condition, Determining $P_{e1}$ and $P_{e2}$ using reduced order $Y_{bus}^{'}$

$$P_{ei} = \Sigma_{j=1}^{j=n} E_i^{'} E_j{}' Y_{ij} cos(\delta_i - \delta_j - \theta_{ij}) \tag{2}$$

where, $n$ represents number of buses and $j, k$ represents the buses. During fault near bus the power injected from generator 1 into the system is zero. Power will be injected to load from generators 2 and and the infinite bus only.

$$\begin{aligned} P_{e1} &= 0 \\ P_{e2} &= |E_2|^2 G_{22} + |E_2||E_3||Y_{23}|cos(\delta_{23} - \theta_{23}) \end{aligned} \tag{3}$$

$\delta$ is the rotor angle, $E_2^{'}$ is the transient internal voltage of generator and $\theta_{jk}$ is the angle of $jk^{th}$ element of $Y_{bus}$ matrix. ($P_c = |E_2^{'}|^2 G_{22}$) and ($P_{max} = |E_2^{'}||E_3||Y_{23}|$) are the

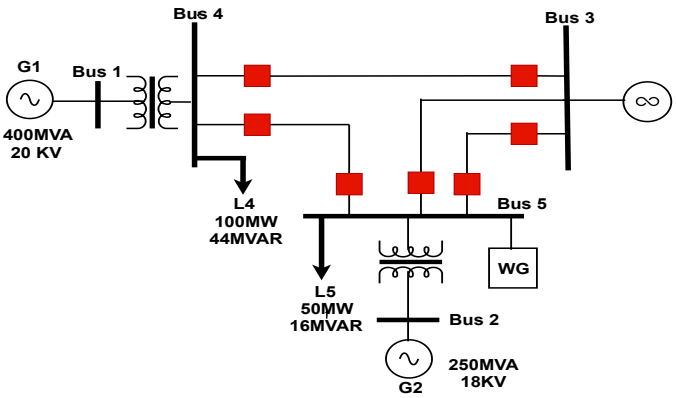



**Fig. 1:** Single line diagram of 5 bus system

constants for the considered network. Bus 3 is a slack bus so $\delta_{23} = \delta_2$ and $\delta_{13} = \delta_1$. Initially value of mechanical power ($P_m$) is same as the electrical power ($P_e$) from the load flow data and the swing equation during the fault is represented as in (4)

$$\frac{d^2\delta_i}{dt^2} = \frac{180f}{H_i}(P_{mi} - P_{ei}) = \frac{180f}{H_i}(P_{ai}) \tag{4}$$

where ($P_{ai}$) is accelerating power of($i_{th}$) generator

### B. Desired swing equation postfault

The fault is cleared by simultaneously opening the circuit breaker at the end of the faulted $4-5$ line. It can be noticed that both the generating bus (1,2) is connected radially to the infinite bus and the desired power swing equation post fault condition is

$$\begin{aligned} P_{e1} &= |E_1^{'}|^2 G_{11} + |E_1^{'}||E_3||Y_{13}|cos(\delta_{13} - \theta_{13}) \\ P_{e2} &= |E_2^{'}|^2 G_{22} + |E_2^{'}||E_3||Y_{23}|cos(\delta_{23} - \theta_{23}) \end{aligned} \tag{5}$$

Electrical power condition obtained in (3) and (5) are used for obtaining the swing equation during fault and the post fault equation after clearing the fault. Using the offline and online simulators, the swing equations are further examined for both the cases for pre-fault, post-fault, and after-breaker operation by checking power, frequency, and bus voltages.

### C. Determination of the critical clearing time

The CCT ($t_{cr}$) [16] is the allowable time for clearing interference and allowing the system to continue functioning normally and in synchronism. The length of time allowed before clearing a fault can be found by determining swing curves for various clearing times, as shown in the simulation. Conventionally, an approximate point-by-point approach [17] is used to determine the CCT. In this method, the swing curve is analyzed for the small time interval of ($\Delta t = 0.05sec$) using (6). Starting from the initial rotor angle, the next position can be calculated.

$$\begin{aligned} \Delta\delta_n &= \delta_{(n-1)} + KP_{a,n-1} \\ P_{a,n-1} &= P_m - P_c - P_{max}sin(\delta_{(n-1)} - \theta) \end{aligned} \tag{6}$$

where, $K = \frac{180f}{H}\Delta t^2$ As the rotor angle $\delta$ is changing continuously, Accelerating power $P_a$ and the speed continuously



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varies. Before the fault, $P_a = 0$, and as the fault occurs $P_a$ attains a non-zero value. However, this discontinuity is measured at each interval and the CCT ($t_{cr}$) is calculated by taking the average interval of two nonzero accelerating power values.

### D. Circuit Breaker Resistance

The closed resistance of a circuit breaker also have a significant impact on transient stability in a power system. When a fault occurs in a power system, the circuit breaker is designed to open and isolate the faulty section of the system. However, during the process of opening, an arc is created across the contacts of the circuit breaker. This arc creates a low-impedance path that can cause a significant increase in the system impedance, leading to a decrease in the fault current and a corresponding increase in the transient recovery voltage [18]. Transient Recovery Voltage (TRV) refers to the voltage that occurs across the contacts of a circuit breaker after it has interrupted a fault [19]. This voltage surge can be harmful to the circuit components due to its rapid rise. This is because the closed resistance of the circuit breaker impacts the rate at which the current in the circuit is interrupted. Conversely, a lower closed resistance will result in a faster rate of current interruption, which can result in a lower TRV. Therefore, it is important to select circuit breakers with appropriate closed resistance values to ensure that transient stability is maintained and have less TRV during fault conditions. Mathematically representing the relationship between TRV and breaker closed resistance is through the following equation:

$$TRV = (L\frac{di}{dt}) + V_c$$
$$R_b = \frac{1}{I_b}(V_c - (\frac{V_f}{\alpha})) \tag{7}$$

where L is the inductance of the system, i is current during the interruption process, and $V_c$ is the voltage across the breaker contacts at the instant of current interruption. $I_b$ is the peak fault current, $R_b$ is the breaker closed resistance, $V_f$ is the voltage drop across the fault, and $\alpha$ is the rate of rise of recovery voltage.

### E. Post fault analysis

The stability of the power system is promoted by quick fault clearing. In fact, speeding up fault clearance is usually the most efficient and cost-effective strategy to enhance the power system's transient stability. The synchronous machines' rotors will accelerate differently after a fault, causing the angle between two equivalent machines to change, which could cause the system to become unstable [20]. Each bus has PMU installed to monitor the voltage and frequency continuously. PMU readings are compared with the reference values at each instant. A three-phase fault with an internal fault resistance of $0.1$ ohm is introduced for a duration of $0.1sec$. Real-time simulation findings have shown that transients and power flow parameters differ from their reference values as the fault occurs. The mechanical power input is immediately surpassed by the electrical power output, causing the accelerating power negatively, following the equal-area criteria [15]. Even if the rotor angle has advanced to a new value, the rotor slows down until it reaches synchronous speed, but it cannot stay there because the accelerating power is still negative. The breaker operation is activated accordingly at $t < t_{cr}$ to attain the steady state of the network.

## III. Simulations tools used for transient stability analysis

A single line flow diagram of the $5$ bus network under test is designed on PWS and RTDS as shown in Fig. 2 and Fig. 3. The parameters of generating units, load, transformer, and transmission lines has been taken from [15] for the power flow analysis. The system base is chosen as $100$ MVA, $230$KV. PMU are deployed near bus $4$ and $5$ for data monitoring. The power system networks demand the load flow simulation study while interconnecting and coordinating with the distributed resources.

### A. PowerWorld Simulator

Power World Simulator is software that can be used offline and allows to interactively simulate high voltage power systems that can run for several minutes to several days [21]. Power World employs the second-order Runge-Kutta integration method for swing analysis. Power world gives simulated line losses, line flows, and bus voltages, bus angles under various operational settings. This non-real-time simulator tracks the power system's steady state [22]. Various features including fault analysis, transient stability, result plotting, and PV curves are available in this software. To produce converged results, Power World employs the Newtown-Raphson method with numerous iterations.

### B. Real Time Digital Simulator

RTDS, a real-time simulator is a package of both hardware and software (RSCAD) to simulate the Electromagnetic Transient (EMT) phenomenon. Real-time simulation means that the power system operates in its own closed loop and the user may interact with the simulation in real-time observing the effect of control action [23]. Real-time simulators use a differential evolution-based technique suitable for nonlinear and non-differentiable optimization problems [24] [25]. Simulations are performed for discrete-time and constant-time steps. The reason for using the real-time simulation tools is that the Model-Based Design (MBD) can be prepared in RTDS using the library of RSCAD, in which various models of generators and other controllers are available which can be used by drag and drop for modeling of the network. Parameters like the inertia of the machine, excitation parameters, and turbine parameters can also be modified in the run time. Instead of costly and risky tests of power systems using physical test benches, real-time simulators are helpful in the testing and monitoring of the equipment before the installation in the network [26]. The RTDS line model is a distributed, frequency-dependent model based on the traveling wave theory [27]. The reduced time step can be achieved by using frequency-dependent line



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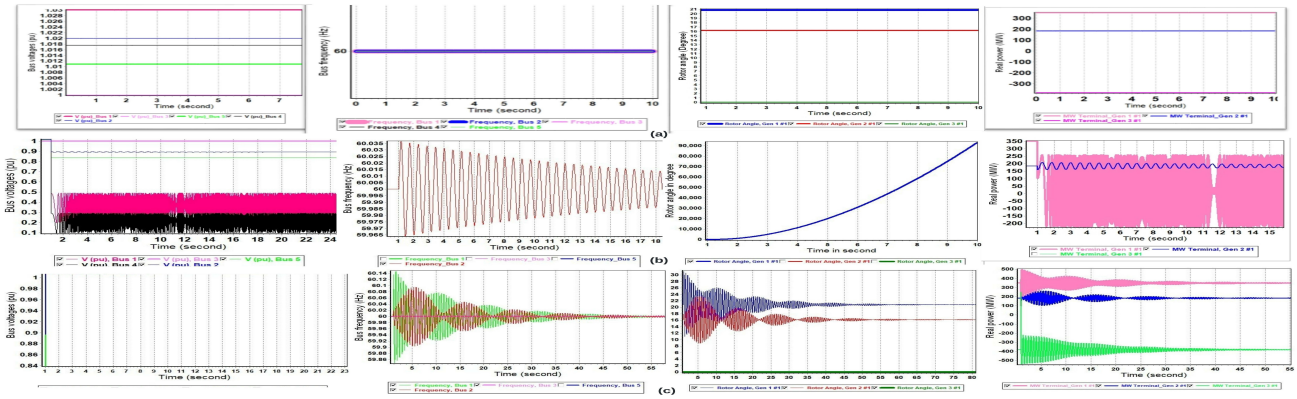


**Fig. 2:** Bus Voltage (V), Bus Frequency (f), Rotor Angle ($\delta$), Generator Real Power (P) using PowerWorld Simulator: (a) Before Fault (b) During Fault near Bus 4 with breaker closed (c) Post Fault With breaker open at $t_{cr} = 0.05sec$

models. Effective frequency bandwidths are thereby enhanced, allowing for more accurate EMT analysis.

For the identification of the faults, micro Phasor Measurement Units (PMUs) are placed nearby buses. Parameters such as voltage, frequency, phase, and Rate of Change of Frequency(ROCOF) are measured for identifying the fault location and decision-making for the opening of the circuit breaker to achieve stability after the transients in the network. PMUs use Global Positioning System (GPS) signals to give real-time synchronised measurements in the power system with more than one microsecond synchronisation precision. The GTNET-PMU firmware is used for applications relating to phase measurement units. Many common network protocols are used by the GTNETx2 card to interface the RTDS to external hardware over a LAN connection.

Results obtained on both RTDS and PWS are tabulated and analyzed graphically in the simulation section to observe the impact of fault on the system.

## IV. Simulation and Discussion

Power swings are examined before, during, and after the fault and breaker operation after the fault to stabilize the network. Load variation impact also has been analyzed. For different case studies, various parameters such as Bus Frequency(f), Bus Voltage(V), and Generator Real Power(P), have been examined. Along with a comparison of non-real-time and real-time simulators, the simulation results are shown graphically (Fig. 2, Fig. 3, Fig. 4, Fig. 6, Fig. 5) and in tabular form for several case studies before and after faults.

### A. Load flow analysis before fault

Under steady-state conditions, the input mechanical power is equivalent to the output electrical power and the generators are running at synchronous speed. Using (1), the rotor angle determined for the generator1 is $20.82°$ and for the generator 2 is $16.19°$ as can be seen graphically in Fig. 2(a). These theoretical values are almost the same as the result obtained in Table I for PWS and RTDS. The load flow analysis using both the simulators is tabulated in Table I, it can be seen that during pre-fault results obtained from both the software are almost the same.

### B. Impact of Three-phase fault near Bus 4

When a fault is introduced the main impact is on voltage, phases, and frequency. These are analyzed using PMU installed in RSCAD. When a fault is introduced near bus 4 the output electrical power of the nearest generator suddenly goes down as seen in Table II and in Fig. 4 while mechanical power is unchanged. The rate of change of the kinetic energy stored in the rotor masses must be accounted for to describe the power difference. Only an increase in speed that results from the constant acceleration power $P_m$ can do this. $P_m$ stays constant until the point of critical clearance, at which the circuit breaker must be triggered to restore system stability.

It is evident from Fig. 4(a) and Fig. 5(a) that the closest interconnections Bus 4 and generator $G_1$ are the most negatively impacted at the moment of fault, with a large dip in voltage at bus 4 and more power decrement in $G_1$ as compared to $G_2$. A minor difference in the monitored value of Table II on both software is due to the fact that RTDS is real-time and the components used for analysis are close to practical aspects, i.e., machine, fault, and circuit breaker resistance are not zero, but have some values, whereas in PWS all the components used are ideal. All of the generators are initially in a steady state, and the speed increases abruptly when the fault is introduced.

### C. Circuit Breaker operation

The circuit breaker is essential every time a fault occurs. Fig.5 shows that if a circuit breaker is not activated during a fault, the voltage at Bus 4 drops to zero and cannot be recovered, while the voltage at Bus 5 settles to a new value that is not similar to the voltage in the base case. Fig.4 demonstrates that the power generation of $G_1$ settles down to zero while power of $G_2$ keeps increasing, which leads to network instability and causes the entire network to lose synchronism. The network's frequency also decreases without an active circuit breaker, as shown in Fig.6. All of this highlights the importance of circuit breakers in clearing faults in a network. The changes in the network's voltage, power, and frequency during transients generate deviations from their nominal values, as illustrated in Fig.4, Fig.6, and Fig.5.



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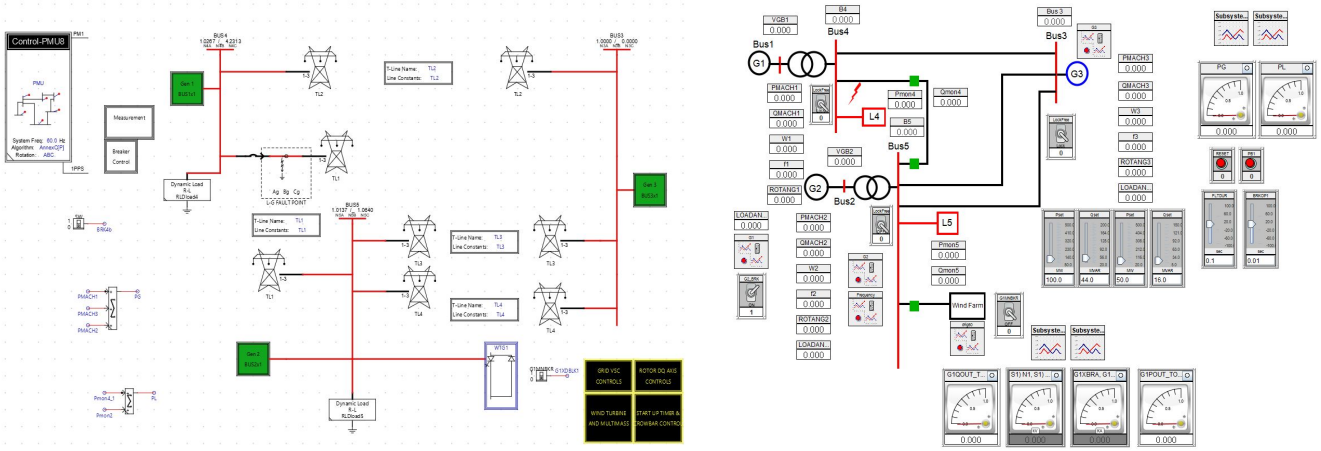


**Fig. 3:** Bus Voltage (V), Bus Frequency (f), Rotor Angle ($\delta$), Generator Real Power (P) using Real time Simulator: (a) During Fault near Bus 4 with breaker closed (b) Post Fault With breaker open at $t_{cr} = 0.05sec$

**TABLE I:** Load flow analysis under steady state using : Real Time Digital Simulator (RTDS) and Offline PowerWorld Simulator

| Bus No. | Bus type | Voltage V (pu) | | Rotor Angle (degree) | | Real power generated (MW) | | Reactive power (MVAr) | |
|---|---|---|---|---|---|---|---|---|---|
| | | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | Power world |
| 1 | PV | 1.03 | 1.03 | 20.81 | 20.82 | 350.4 | 350 | 71.281 | 71.25 |
| 2 | PV | 1.02 | 1.02 | 17.2 | 16.39 | 185.28 | 185 | 29.799 | 29.80 |
| 3 | Slack | 1.00 | 1.00 | 0.00 | 0.00 | -372.50 | -381 | -26.59 | -27.00 |
| 4 | PQ | 1.016 | 1.0175 | - | - | 99.58 | 99.8 | 43.89 | 43.83 |
| 5 | PQ | 1.015 | 1.0109 | - | - | 49.46 | 50.5 | 16.02 | 16.24 |

**TABLE II:** During Three phase Fault near Bus 4 with closed circuit breaker

| Fault on Transmission line 4-5 near bus 4 | | | | | | | | | | Fault on bus 5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bus | | V (pu) | | Rotor angle | | P (MW) | | f(Hz) | | V (pu) | | Rotor angle | | P (MW) | | f(Hz) | |
| | | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS |
| 1 | PV | 0.5 | 0.49 | nd | nd | 110 | 115 | 61.1 | 62.0 | 0.78 | 0.876 | 32 | 34.5 | 536 | 498 | 59.7 | 60.01 |
| 2 | PV | 0.89 | 0.86 | 9.6 | 9.75 | 220 | 210 | 59.4 | 60.3 | 0.54 | 0.304 | nd | nd | 181 | 0 | 60.1 | 61.00 |
| 3 | Slack | 1.00 | 1.00 | 0.00 | 0.00 | -200.06 | -198 | 59.47 | 60.0 | 1.00 | 1.00 | 0.00 | 0.00 | 579 | 580 | 59.34 | 60.06 |
| 4 | PQ | 0.24 | 0.25 | - | - | 103 | 100 | 61 | 60.3 | 0.85 | 0.816 | - | - | 150 | 100 | 60.47 | 60.06 |
| 5 | PQ | 0.99 | 1.01 | - | - | 48.04 | 50 | 59.7 | 59.9 | 0.28 | 0.00 | - | - | 80 | 50 | 59.45 | 60.13 |

**TABLE III:** Impact of breaker operation at $t < t_{cr}$ after the fault clearance

| Post fault of fault on transmission line 4-5 near bus 4 with tcr=0.01 sec | | | | | | | | | | Fault on bus 5 with tcr=0.05 sec | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2*Bus | | V (pu) | | Rotor angle | | P (MW) | | f(Hz) | | V (pu) | | Rotor angle | | P (MW) | | f(Hz) | |
| | | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS | RTDS | PWS |
| 1 | PV | 1.03 | 1.03 | 21.2 | 20.84 | 351.4 | 350.0 | 60 | 60 | 1.03 | 1.03 | 19.9 | 20.8 | 358 | 350 | 60 | 60 |
| 2 | PV | 1.02 | 1.02 | 17.4 | 16.19 | 187 | 185.0 | 60 | 60 | 1.02 | 1.02 | 18.8 | 16.2 | 189 | 185 | 60 | 60 |
| 3 | Slack | 1.00 | 1.00 | 0.00 | 0.00 | -373 | -380.5 | 60 | 60 | 1.00 | 1.00 | 0.00 | 0.00 | -400 | -380 | 60 | 60 |
| 4 | PQ | 1.018 | 1.018 | - | - | 100 | 100.0 | 60 | 60 | 1.018 | 1.018 | - | - | 100 | 100 | 60 | 60 |
| 5 | PQ | 1.011 | 1.011 | - | - | 50 | 50.0 | 60 | 60 | 1.011 | 1.011 | - | - | 50 | 50 | 60 | 60 |

However, with the help of a circuit breaker, the network can retain its synchronism after some oscillation without sustaining significant damage.

### *D. Critical Clearing Time ($t_{cr}$) and Breaker Resitance*

Designing of circuit breaker and relays need the information about $tcr$, so that the total time taken by them to clear fault should be less than the $tcr$ for stable operation of the system. Thus for this purpose various parameters such as frequency, voltages are monitored for various clearing time and we obtain an optimum $tcr$ by which network doesn't violates its stability limits and remains in synchronism.

Fig. 7, displays the bus 4's frequency and voltage for different clearing time. According to standards, the permissible range for grid frequency is 3 % of nominal i.e. 58.2 Hz to 61.8 Hz and similarly for voltage stability the range is 5 % of nominal i.e. 0.95 pu to 1.05 pu. Fig. 7 shows that the network is beyond the stability constraints as the time

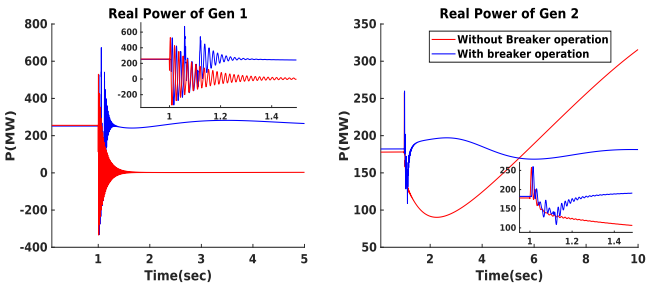



**Fig. 4:** Power(MW) under fault conditions with and without the operation of a breaker: (a) Gen 1 (b) Gen 2

required to clear the fault increases, which could lead to major network instability. The frequency and voltage deviation likewise increase as the $t_{cr}$ rises. This indicates an unstable mode for the system. If a longer time is needed to clear the fault, the system will lose synchronization, hence the $t_{cr}$ needs to be minimal to keep the system in synchronization [28]. So



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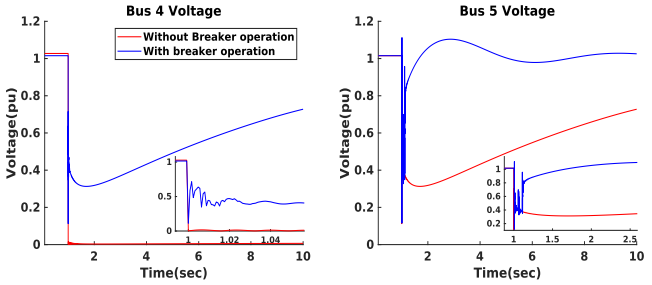



**Fig. 5:** Voltage(pu) under fault conditions with and without the operation of a breaker: (a) Bus 4 (b) Bus 5

for this network obtained value from Fig. 7 is $tcr = 0.01sec$. Circuit breakers can be designed with clearing time less than or equal to $0.01sec$.

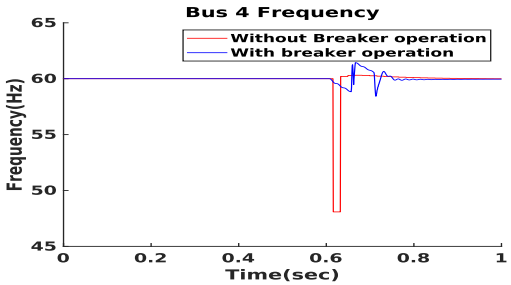



**Fig. 6:** Frequency(Hz) of bus 4 under fault conditions with and without the operation of a breaker

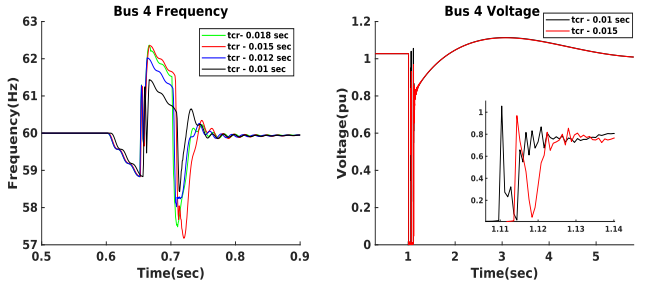



**Fig. 7:** Impact of various fault clearing times on: (a) Bus Frequency(Hz) (b) Bus Voltage(pu) of Bus 4

Fig. 8 shows that an increase in breaker closed resistance will result in an increase in the voltage across the breaker contacts and the current interruption time increases. This, in turn, will increase the magnitude of the TRV, assuming all other factors remain constant. For this particular network, a breaker resistance of 1 micro ohm has been selected in order to the voltage transients and the current interruption time.

### *E. Impact of Wind Generator Integration during Load Variation*

This section analyzes the effect of load variation on the voltage and frequency of the network. The alternator's speed and supply frequency depends on the operational power requirement, which the alternator's excitation determines. Reactive power demand is also directly related to these factors. When

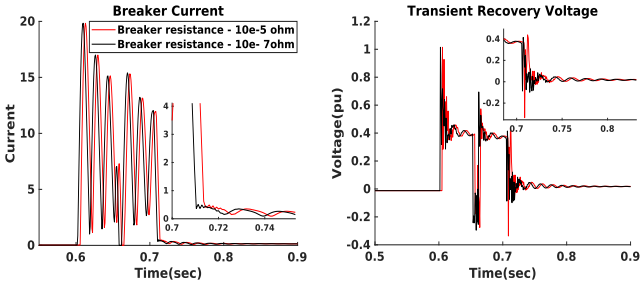



**Fig. 8:** Impact of Breaker Resistance on : (a) Breaker interruption current (b) TRV(pu)

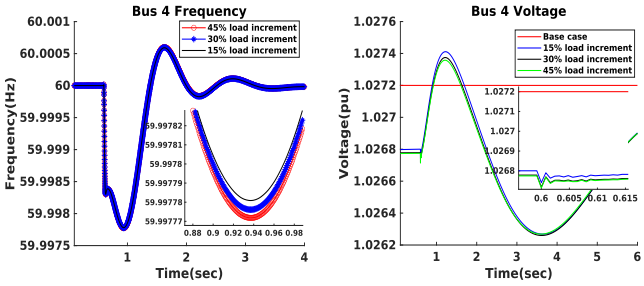



**Fig. 9:** Load variation impact on bus 4 due to 15 %, 30 % and 45 % increment at bus 5: (a) Frequency (Hz) (b) Voltage (pu)

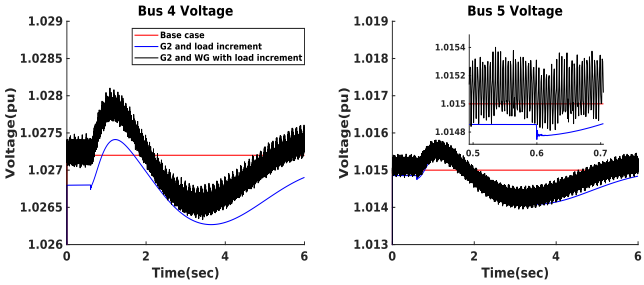



**Fig. 10:** Impact of of Wind Generation(WG) integration on: (a) Bus 4 voltage(pu) (b) Bus 5 voltage(pu)

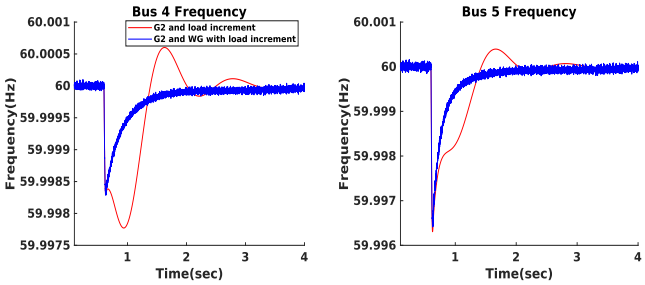



**Fig. 11:** Impact of of Wind Generation(WG) integration on: (a) Bus 4 Frequency(Hz) (b) Bus 5 Frequency(Hz)

the active power demand increases by 15%, 30%, and 45% at bus 5, the rotor's kinetic energy is utilized to meet the demand, slowing down the rotational speed and decreasing the frequency, as shown in Fig.9(a). As the load increases, the frequency dip becomes more profound. Fig.9(b) shows the voltage variation with load increment; as the load increases, the voltage drops from the base value. After some time, both frequency and voltage eventually settle down to their base val-



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ues. However, if the load increases significantly, the network may lose synchronization. To maintain synchronization, wind turbines, capacitors, or other compensating devices are needed to balance the demand.

The wind turbine is preferably connected at the strongest bus, as suggested in [29]. The two PQ buses (Buses 4 and 5) are comparably strong and have almost the same voltage level, according to the network's base case voltage profile. The voltage levels on Buses 4 and 5 are nearly 1.016 p.u. and 1.015 p.u., respectively. The simulation is performed with 15% Load increment in the network. From Fig. 10 improvement in voltage profile has been observed with the integration of the wind farm at bus 5. In a same manner, in Fig. 11 frequency improvement can be seen after the integration of WG.

## V. CONCLUSION

This study emphasizes the importance of considering the impact of fault clearance, circuit breaker closed resistance , wind generation, and load increment on the transient stability of power networks.

1) A transient stability analysis of the 5-bus network has been conducted by simulating the effects of a three-phase fault and its subsequent clearing in RTDS.
2) The simulation results indicates that the circuit breaker resistance and the critical clearing time has an impact on the system's transient recovery voltage to regain the stability.
3) Integrating the wind farm into the network improves the system's stability and reduces the effects of load increment by providing a voltage and the frequency support.

This study can serve as a basis for further research into improving the transient stability of power systems and developing more reliable and robust power systems that ensures the stable and secure operation under different operating conditions.

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